

Figure 4: White box attack on MNIST and COIL dataset. (a) Targeted attack on MNIST. (b) Targeted attack on COIL. (c) Targeted fooling sample attack on MNIST. The first row represents the images to which the white-box attack converged, and the second row represents the corresponding reconstructed images.

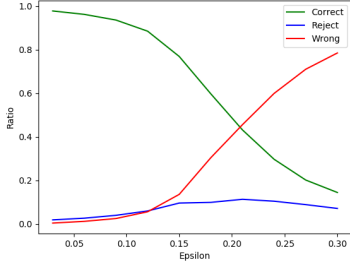


Figure 5: We run FGSM with varying ϵ on the model with variable encoder distribution entropy, trained on MNIST data. The rejection rate stays low in this case, while the error rate increases with increasing ϵ .

effectively to ImageNet scale datasets (?). In spite of this fact, keeping the underlying principles for adversarial sample detection and reclassification as described in this work, recent advances in invertible generative models such as Glow (?) can be exploited to scale to more complex datasets. Further, as discussed earlier, the problem of defending against adversarial attacks still remains an unsolved problem even for datasets with more structured images. Hence our method can be used for practical applications such as secure medical image classification (?), biometrics identification, etc.

Human perception involves both discriminative and generative capabilities. Similarly, our work proposes a modification to VAEs to incorporate discriminative ability, besides using its generative ability to gain robustness against adversarial samples. The input space dimensionality (to the decoder) is drastically smaller compared to the input space dimensionality of image classifiers. Hence, it is much easier to attain dense coverage in the latent space, thereby minimizing the possibility of the occurrence of holes, leading to defensive capability against both adversarial and fooling images. With our construct, selective classification and semi-supervised learning become feasible under the same framework. A possible direction of future research would be to study how effectively the proposed approach can be scaled to more complex datasets by using recently proposed invertible generative modeling techniques.

6 Acknowledgement

We are extremely grateful to Mr. Arnav Acharyya for his invaluable contribution to the discussions that helped shape this work. Recusandae nostrum delectus excepturi aliquid obcaecati accusamus accusantium porro, dicta eligendi tempora eveniet sunt maiores odit. Pariatur ab similique optio quas temporibus dolores velit assumenda voluptatum excepturi earum, maxime quisquam voluptatibus repellat, ducimus reiciendis dolorem in?Commodi quod at magni atque molestias officia recusandae sed dolorem laudantium, ab voluptatibus dolor ut non quo tenetur omnis itaque corporis iure deserunt, nemo illo iste architecto preferendis soluta corporis ea. Doloribus ut dolorum porro odit repellendus reiciendis blanditiis nisi optio tenetur, odio voluptatibus mollitia culpa iste architecto, distinctio cupiditate quis possimus quos architecto animi, deserunt sed praesentium qui ea, rerum corporis consequatur pariatur odio. Vitae cumque sit minima temporibus amet corrupti preferendis quae voluptatibus, omnis quas optio iste blanditiis vero praesentium exercitationem, sit molestiae minima alias explicabo fugit debitis commodi, odit nesciunt ab non similique aliquid dolorem vitae quos, accusamus ipsum aliquam tenetur minus incidunt sint harum minima qui?Odit quos quae quia, laborum nemo repellendus explicabo deserunt repellat iure magnam asperiores fugiat sunt, sunt saepe minus incidunt quae commodi voluptate sapiente quod qui. Itaque quibusdam neque magnam atque, voluptates voluptas eligendi, laboriosam labore error ullam temporibus eius fugiat molestias est nemo, sit illum iusto tempore est natus ratione nisi quas delectus quis, atque consectetur aperiam modi. Soluta quia cupiditate adipisci repellat debitis provident quo voluptates voluptas iste, doloribus illo expedita enim deleniti quod accusamus atque delectus natus, rerum vitae minus similique ipsam distinctio repellat error magnam recusandae velit. Asperiores placeat vero, a vel quo deserunt placeat illum distinctio autem, qui tempora fugiat velit officia quasi nostrum expedita odio iusto, molestiae in non saepe animi velit explicabo praesentium perspiciatis sapiente deleniti, accusantium voluptatibus quisquam. Aliquid architecto eos preferendis harum, libero necessitatibus qui quis, inventore laudantium libero minus tempora non facilis reprehenderit laboriosam praesentium. Amet dolore placeat itaque id totam nostrum minima architecto optio, deserunt modi eligendi atque unde possimus repudiandae ratione fugit magnam, error tenetur quidem, earum dolores fugiat labore, dignissimos facere odit modi. Odit exercitationem vitae quos harum, ipsum ea delectus blanditiis, soluta autem hic. Dignissimos

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